Software Test Plan (STP)

Project Detail	Value
Project Title	Smart Complaint Management System for a Community
Group	Team 9
Course	UE24CS341A – Software Engineering
Methodology	Agile
Team Lead	Riya Saju Vithayathil (PES1UG24CS378)
Team Members	Riya Saju Vithayathil; Ritu Ranganath Kashyap; Rohan Suresh; Rohit Krishna R
Version	1.0

1. Introduction

1.1 Purpose

This Software Test Plan defines the testing strategy, test scenarios, test cases, security testing and requirement traceability for the Smart Complaint Management System. It is based on the submitted SRS and covers the approved in-scope functionality.

1.2 Scope

Testing covers user registration/authentication, complaint registration, categorisation, assignment, processing, tracking, search/filtering, notifications, resolution/closure, feedback/rating, administration and reporting. Out-of-scope enhancements listed in the SRS are not part of this test plan.

1.3 References

- Software Engineering Level 3 Student Submission Guide – Deliverables
- Software Requirements Specification template
- Smart Complaint Management System – SRS / Project Synopsis.

1.4 Definitions

- SCMS – Smart Complaint Management System
- SRS – Software Requirements Specification
- SAD – Software Architecture & Design Specification
- STP – Software Test Plan
- RBAC – Role-Based Access Control.

2. Test Items

- Auth/User Module
- Complaint Module
- Category/Assignment Module
- Processing/Status Module
- Search/Tracking Module
- Notification Module
- Feedback Module
- Admin/Reporting Module
- Database (MySQL)

3. Features to be Tested

- FR-019: Notifications for complaint lifecycle events
- FR-021: Resident feedback after resolution
- FR-022: Feedback and rating linked to the complaint
- FR-023: Administrator management of resident/staff accounts
- FR-024: Administrator dashboard of counts and statuses
- FR-025: Basic reports/statistics
- FR-001–FR-018 and FR-020: registration/authentication, complaint registration, categorisation, assignment, processing, tracking, search/filtering, resolution/closure.
- NFR-001–NFR-006: performance, operational reliability, usability, reliability, maintainability and testability.
- SEC-001–SEC-005: password protection, role-based access control, input validation, session security and attachment security.

4. Features Not to be Tested

Out-of-scope AI classification, AI priority prediction, SMS/WhatsApp integration, native mobile applications, advanced mapping, predictive analytics and large-scale multi-organisation deployment are not tested.

5. Test Approach / Strategy

Section 4: Test Strategy & Execution Approach: Defines unit, integration, functional, usability, performance, and security testing phases across local, staging, and production environments.

Section 5: Environmental Requirements: Staging server configured with Spring Boot backend, MySQL 8.0 DB, and mock notification services.

Section 6: Entry & Exit Criteria: Entry: 100% compilation, design sign-off, unit tests passing. Exit: 0 Critical/High open defects, 100% execution of planned test cases.

Section 7: Defect Management: Defect tracking via Jira using standard severities (Blocker, Critical, Major, Minor).

Section 8: Test Deliverables: Test Cases, Execution Logs, Bug Reports, Traceability Matrix, Test Summary Report.

Section 12: Responsibilities: P3 owns test case creation and execution for processing, non-functional performance, usability, and reliability tests.

Section 14: Risks & Assumptions: Staging environment mirrors production capacity; API rate limits are configured for test runs.

5.1 Security Validation

Security validation will verify that authentication, authorization, input validation, session protection and complaint attachment controls satisfy the security requirements defined in the SRS.

SEC-001 – Password Protection Verify that passwords are not stored in plaintext and that stored credentials use the configured secure password representation.

SEC-002 – Role-Based Access Control Verify that residents cannot access staff or administrator functionality and that protected operations enforce role permissions.

SEC-003 – Input Validation Submit malformed, unexpected or invalid input through registration and complaint forms and verify that the system validates and safely handles the input.

SEC-004 – Session Security Attempt to access protected pages or APIs using invalid, expired or missing authentication sessions and verify that access is rejected.

SEC-005 – Attachment Security Attempt to upload supported and unsupported file types and files exceeding the configured size restriction. Verify that only permitted attachments are accepted.

Security testing will include both valid and invalid scenarios. High-priority defects affecting authentication, authorization or protected data access should be treated as release-blocking issues.

6. Test Environment

Item	Planned Environment
Client	Web browser on a standard student/development machine.
Backend	Python Flask application.
Database	MySQL test database.
Testing	pytest for automated tests; Postman for API testing where applicable.
Version Control	Git/GitHub.
CI/CD	Jenkins, as planned in the project tool stack.
Containerization	Docker, as planned for deployment.
Static Analysis	SonarQube or similar, as planned.
Design/Test Evidence	Screenshots, logs, test results and traceability records.

7. Test Schedule

8. Test Deliverables

Section 8: Test Deliverables: Test Cases, Execution Logs, Bug Reports, Traceability Matrix, Test Summary Report.

9. Roles and Responsibilities

Section 12: Responsibilities: P3 owns test case creation and execution for processing, non-functional performance, usability, and reliability tests.

10. Risks and Mitigation

Risk	Mitigation
These tests need complaints in specific statuses produced by other modules.	A seed script loads complaints in every status into the test database.

11. Assumptions & Dependencies

No separate assumptions/dependencies text is provided in the supplied contribution documents.

12. Suspension & Resumption Criteria

No separate suspension/resumption criteria are provided in the supplied contribution documents.

13. Test Case Management & Traceability

13.1 Test Cases – Auth/User Module

Test Case ID	Requirement ID	Description	Preconditions	Steps	Expected Result	Type
TC-AUTH-01	FR-001	Register with valid details	Registration page available	Enter all required valid details and submit	Resident account is created successfully	Positive
TC-AUTH-02	FR-001	Register with already registered account details	Registration page available	Enter an email address that is already registered and submit the form	Account creation is rejected with appropriate validation message	Negative
TC-AUTH-03	FR-002	Registration with missing required field is rejected	Registration page available	Leave one required field empty and submit the form	Registration is rejected and an appropriate validation message is displayed	Positive
TC-AUTH-04	FR-002	Registration with invalid information is rejected	Registration page available	Enter invalid information, such as an incorrectly formatted email address, and submit the form	Registration is rejected and an appropriate validation message is displayed	Negative
TC-AUTH-05	FR-003	Login with valid credentials	Registered resident exists	Enter valid email/password and login	Authenticated session is established	Positive
TC-AUTH-06	FR-003	Login with incorrect password	Registered resident exists	Enter valid email and incorrect password	Login is rejected	Negative
TC-AUTH-07	FR-004	Access permitted resident functionality	Resident successfully authenticated	Access resident dashboard/complaint functionality	Permitted functionality is accessible	Positive
TC-AUTH-08	FR-004	Resident attempts staff/admin function	Resident successfully authenticated	Attempt to access protected staff/admin function	Access is denied	Negative

13.2 Test Cases – Complaint Module

Test Case ID	Requirement ID	Description	Preconditions	Steps	Expected Result	Type
TC-CMP-01	FR-005	Submit valid complaint	Resident authenticated	Enter valid complaint details and submit	Complaint is successfully created	Positive
TC-CMP-02	FR-005	Unauthenticated user attempts complaint submission	User not authenticated	Attempt to submit complaint	Submission is rejected and authentication is required	Negative
TC-CMP-03	FR-006	Submit complaint with all mandatory fields	Resident authenticated	Enter title, category, description and location	Complaint is accepted	Positive

TC-CMP-04	FR-006	Submit complaint with missing mandatory field	Resident authenticated	Leave one mandatory field empty and submit	Submission is rejected with validation message	Negative
TC-CMP-05	FR-007	Upload permitted attachment	Resident authenticated	Attach a supported file within configured size limit	Attachment is accepted	Positive
TC-CMP-06	FR-007	Upload unsupported/oversized attachment	Resident authenticated	Attach unsupported or oversized file	Attachment is rejected	Negative
TC-CMP-07	FR-008	Generate Complaint ID after successful submission	Resident authenticated	Submit valid complaint	System generates a unique Complaint ID	Positive
TC-CMP-08	FR-008	Verify unique Complaint IDs are generated	Existing complaints available	Submit multiple valid complaints	Each successfully submitted complaint receives a different Complaint ID	Positive

13.3 Test Cases – Security

Test Case ID	Requirement ID	Description	Preconditions	Steps	Expected Result	Type
TC-SEC-01	SEC-001	Verify secure password storage	Registered user exists	Inspect stored credential representation	Password is not stored in plaintext	Positive
TC-SEC-02	SEC-001	Verify plaintext password is not stored in the database	Registered user exists	Inspect the database record for the registered user's password	The password is stored only in a secure hashed representation and is not readable as plaintext	Negative
TC-SEC-03	SEC-002	Resident accesses permitted functionality	Resident authenticated	Access resident functions	Resident functionality is allowed	Positive
TC-SEC-04	SEC-002	Resident attempts privileged functionality	Resident authenticated	Attempt staff/admin operation	Access is denied	Negative
TC-SEC-05	SEC-003	Submit valid input	Registration/complaint form available	Enter valid expected input	Input is accepted and processed	Positive
TC-SEC-06	SEC-003	Submit malformed/unexpected input	Registration/complaint form available	Enter malformed or unsafe input	Input is rejected or safely handled	Negative
TC-SEC-07	SEC-004	Access protected page with valid session	User authenticated	Access protected functionality	Request is accepted	Positive

TC-SEC-08	SEC-004	Access protected page with expired/invalid session	User session expired/invalid	Attempt protected request	Request is rejected	Negative
TC-SEC-09	SEC-005	Upload permitted attachment	Resident authenticated	Upload supported file within configured size	Attachment is accepted	Positive
TC-SEC-10	SEC-005	Upload prohibited attachment	Resident authenticated	Upload unsupported/oversized file	Attachment is rejected	Negative

13.4 Test Cases – Processing/Status Module

Test Case ID	Requirement ID	Description	Preconditions	Steps	Expected Result	Type
TC-PRC-01	FR-012	View complaints assigned to staff	Maintenance staff logged in; complaints assigned	1. Send GET /api/v1/staff/complaints/assigned	List of assigned complaints is displayed correctly.	Positive
TC-PRC-02	FR-012	View assigned complaints with unauthorized user	Resident user logged in	1. Send GET /api/v1/staff/complaints/assigned	HTTP 403 Forbidden returned; access denied.	Negative
TC-PRC-03	FR-013	Update complaint status using permitted state transition	Complaint in Assigned status	1. Send PATCH /status with status In Progress	Status updated to In Progress.	Positive
TC-PRC-04	FR-013	Update complaint status using illegal state transition	Complaint in Submitted status	1. Send PATCH /status with status Resolved	HTTP 400 Bad Request; state transition rejected.	Negative
TC-PRC-05	FR-014	Add remarks and resolution details upon completion	Complaint in In Progress status	1. Send PATCH /status with status Resolved and resolution remarks	Complaint state set to Resolved with saved remarks.	Positive
TC-PRC-06	FR-014	Resolve complaint without providing resolution details	Complaint in In Progress status	1. Send PATCH /status with status Resolved and empty remarks	HTTP 400 Bad Request; validation failure for missing details.	Negative
TC-PRC-07	FR-020	Close a resolved complaint via proper workflow	Complaint in Resolved status	1. Send PATCH /close request	Complaint state updated to Closed.	Positive
TC-PRC-08	FR-020	Attempt to close an unresolved complaint	Complaint in In Progress status	1. Send PATCH /close request	HTTP 400 Bad Request; complaint	Negative

					cannot be closed.	

13.5 Test Cases – Performance and Non-Functional Requirements

Test Case ID	Requirement ID	Description	Preconditions	Steps	Expected Result	Type
TC-PERF-01	NFR-001	Verify response time under normal test load	Benchmark load running on staging	1. Send 100 concurrent complaint status requests	95% of requests complete within 10 seconds.	Positive
TC-PERF-02	NFR-001	Verify system performance under extreme overload	Load 3x higher than peak load limit	1. Execute stress test suite	System returns 429 Rate Limit/degrades gracefully without crashing.	Negative
TC-NFR-01	NFR-002	Prevent duplicate complaints on retry	Network timeout during submission	1. Resident retries complaint submission	Duplicate record is prevented via idempotency key.	Positive
TC-NFR-02	NFR-003	Usability walkthrough for complaint submission	Resident logged in	1. Complete complaint workflow	Submission completed within ≤ 5 primary screens.	Positive
TC-NFR-03	NFR-004	Creation retry after temporary failure	Simulated server disconnection	1. Submit complaint, drop network, click retry	Single complaint record created upon recovery.	Positive
TC-NFR-04	NFR-005	Core business logic unit test coverage	Development build environment	1. Run unit test suite for core services	Unit tests pass independently with module isolation.	Positive
TC-NFR-05	NFR-006	Test core components without full system	Standalone component setup	1. Execute isolated service tests using test mocks	Service logic functions independently of full deployment.	Positive

13.6 Test Cases – Notification Module

Test Case ID	Requirement ID	Description	Preconditions	Steps	Expected Result	Type
TC-NTF-01	FR-019	Notification on lifecycle events	Complaint in Submitted status	Administrator assigns the complaint; staff updates its status	Resident and assigned staff receive an unread notification for each event	Positive
TC-NTF-02	FR-019	Access another user's notification	Resident A has a notification; resident B authenticated	B requests A's notification	Access is denied; not shown in B's list	Negative

13.7 Test Cases – Feedback Module

Test Case ID	Requirement ID	Description	Preconditions	Steps	Expected Result	Type
TC-FBK-01	FR-021	Submit feedback on resolved complaint	Resident owns a Resolved complaint	Select rating 4, enter comment and submit	Feedback is recorded and shown read-only	Positive
TC-FBK-02	FR-021	Feedback before resolution	Resident owns an In Progress complaint	Attempt to submit feedback	Feedback is rejected; nothing stored	Negative
TC-FBK-03	FR-022	Feedback linked to complaint	Feedback exists for a complaint	Resident and Administrator open the complaint	Both see the same rating on that complaint only	Positive
TC-FBK-04	FR-022	Duplicate feedback	Feedback exists for a complaint	Owner submits feedback again	Second feedback is rejected; original unchanged	Negative

13.8 Test Cases – Admin/Reporting Module

Test Case ID	Requirement ID	Description	Preconditions	Steps	Expected Result	Type
TC-ADM-01	FR-023	Create staff account	Administrator authenticated; department exists	Add staff with valid details	Account is created; new staff can log in	Positive
TC-ADM-02	FR-023	Non-admin attempts account management	Resident or staff authenticated	Access user management	Access is denied	Negative
TC-ADM-03	FR-024	Dashboard counts	Known test data loaded	Administrator opens the dashboard	Total and each status count match the data	Positive
TC-ADM-04	FR-024	Non-admin opens dashboard	Resident authenticated	Access the admin dashboard	Access is denied	Negative
TC-ADM-05	FR-025	Generate reports	Known test data loaded	Run category, period, pending and workload reports	Counts match the data; period respects the date range	Positive
TC-ADM-06	FR-025	Invalid report request	Administrator authenticated	Request unknown type; from later than to	Each request is rejected	Negative

13.9 Requirement Traceability

Requirement ID	Component	Test Case IDs
FR-001	Auth/User Module	TC-AUTH-01, TC-AUTH-02
FR-002	Auth/User Module	TC-AUTH-03, TC-AUTH-04
FR-003	Auth/User Module	TC-AUTH-05, TC-AUTH-06
FR-004	Auth/User Module	TC-AUTH-07, TC-AUTH-08
FR-005	Complaint Module	TC-CMP-01, TC-CMP-02
FR-006	Complaint Module	TC-CMP-03, TC-CMP-04
FR-007	Complaint Module	TC-CMP-05, TC-CMP-06

FR-008	Complaint Module	TC-CMP-07, TC-CMP-08
FR-012	Processing/Status Module	TC-PRC-01, TC-PRC-02
FR-013	Processing/Status Module	TC-PRC-03, TC-PRC-04
FR-014	Processing/Status Module	TC-PRC-05, TC-PRC-06
FR-020	Processing/Status Module	TC-PRC-07, TC-PRC-08
NFR-001	Performance	TC-PERF-01, TC-PERF-02
NFR-002	Operational / Reliability	TC-NFR-01
NFR-003	Usability	TC-NFR-02
NFR-004	Reliability	TC-NFR-03
NFR-005	Maintainability	TC-NFR-04
NFR-006	Testability	TC-NFR-05
FR-019	Notification Module	TC-NTF-01, TC-NTF-02
FR-021	Feedback Module	TC-FBK-01, TC-FBK-02
FR-022	Feedback Module	TC-FBK-03, TC-FBK-04
FR-023	Admin/Reporting Module	TC-ADM-01, TC-ADM-02
FR-024	Admin/Reporting Module	TC-ADM-03, TC-ADM-04
FR-025	Admin/Reporting Module	TC-ADM-05, TC-ADM-06
SEC-001	Auth/User Module	TC-SEC-01, TC-SEC-02
SEC-002	Auth/User Module	TC-SEC-03, TC-SEC-04
SEC-003	Auth/User + Complaint Module	TC-SEC-05, TC-SEC-06
SEC-004	Auth/User Module	TC-SEC-07, TC-SEC-08
SEC-005	Complaint Module	TC-SEC-09, TC-SEC-10

13.10 Test Data / Evidence to Collect

- Registration screenshots and automated test output for valid and invalid registration.
- Authentication evidence for successful and failed login attempts.
- Authorization evidence showing resident denial for staff/administrator functions.
- Evidence that passwords are not stored in plaintext.
- Input validation evidence for malformed registration and complaint data.
- Session-security evidence for invalid or expired sessions.
- Attachment validation evidence for permitted, unsupported and oversized files.
- Evidence of unique Complaint IDs for successfully submitted complaints.
- Notification list screenshots after assignment and status change.
- Evidence that a user cannot access another user's notifications.
- Feedback submission evidence for valid, early and duplicate attempts.
- Evidence that stored feedback is linked to the correct complaint.
- Authorization evidence showing resident/staff denial for administrator functions.
- Staff account creation evidence.
- Dashboard and report output compared against the seeded test data.

14. Test Metrics & Reporting

Section 14: Risks & Assumptions: Staging environment mirrors production capacity; API rate limits are configured for test runs.